

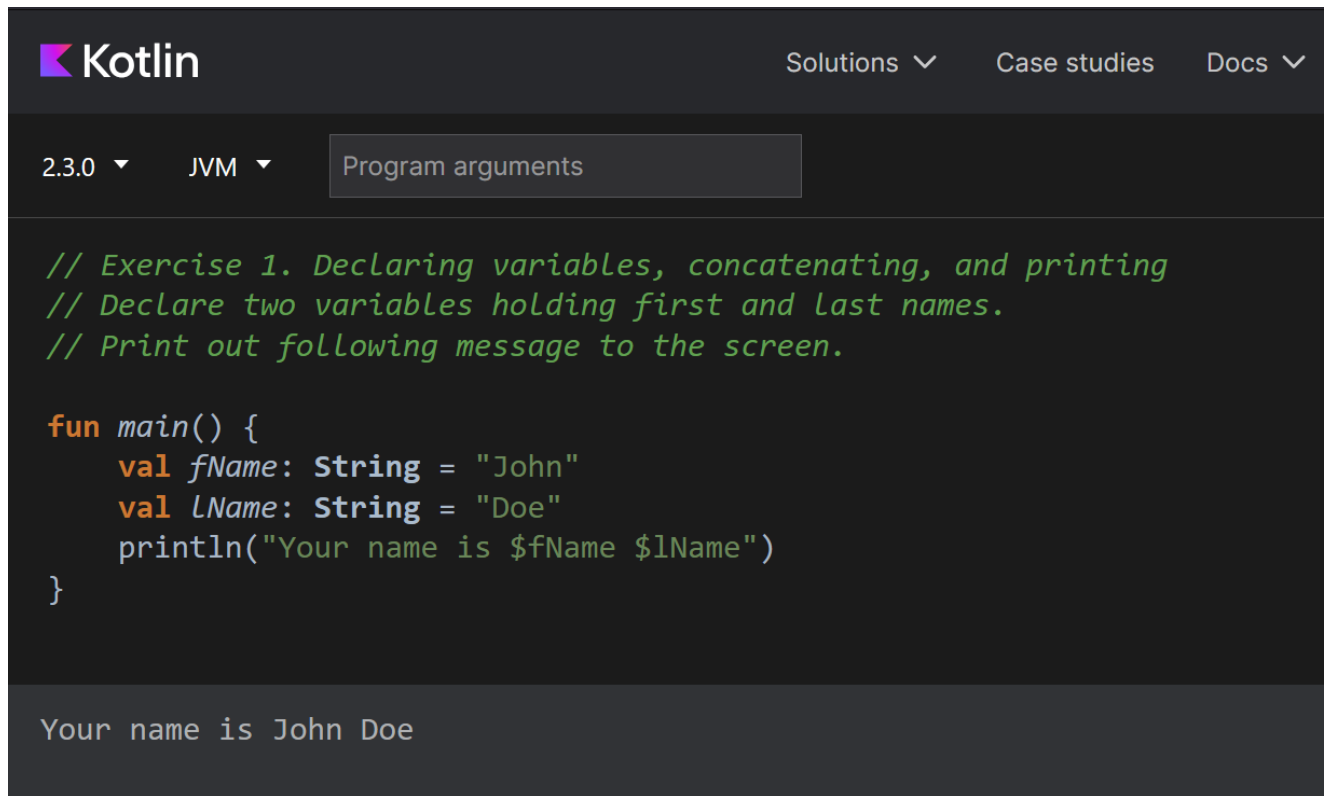
# Exercises. Basics of Kotlin programming language

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## Exercise 1



The screenshot shows the Kotlin IDE interface. At the top, there's a header with the Kotlin logo and navigation links: "Solutions", "Case studies", and "Docs". Below the header, there are dropdown menus for "2.3.0" and "JVM", and a text input field containing "Program arguments". The main area displays Kotlin code for Exercise 1, which declares two variables, concatenates them, and prints the result. The output at the bottom shows "Your name is John Doe".

```
// Exercise 1. Declaring variables, concatenating, and printing
// Declare two variables holding first and last names.
// Print out following message to the screen.

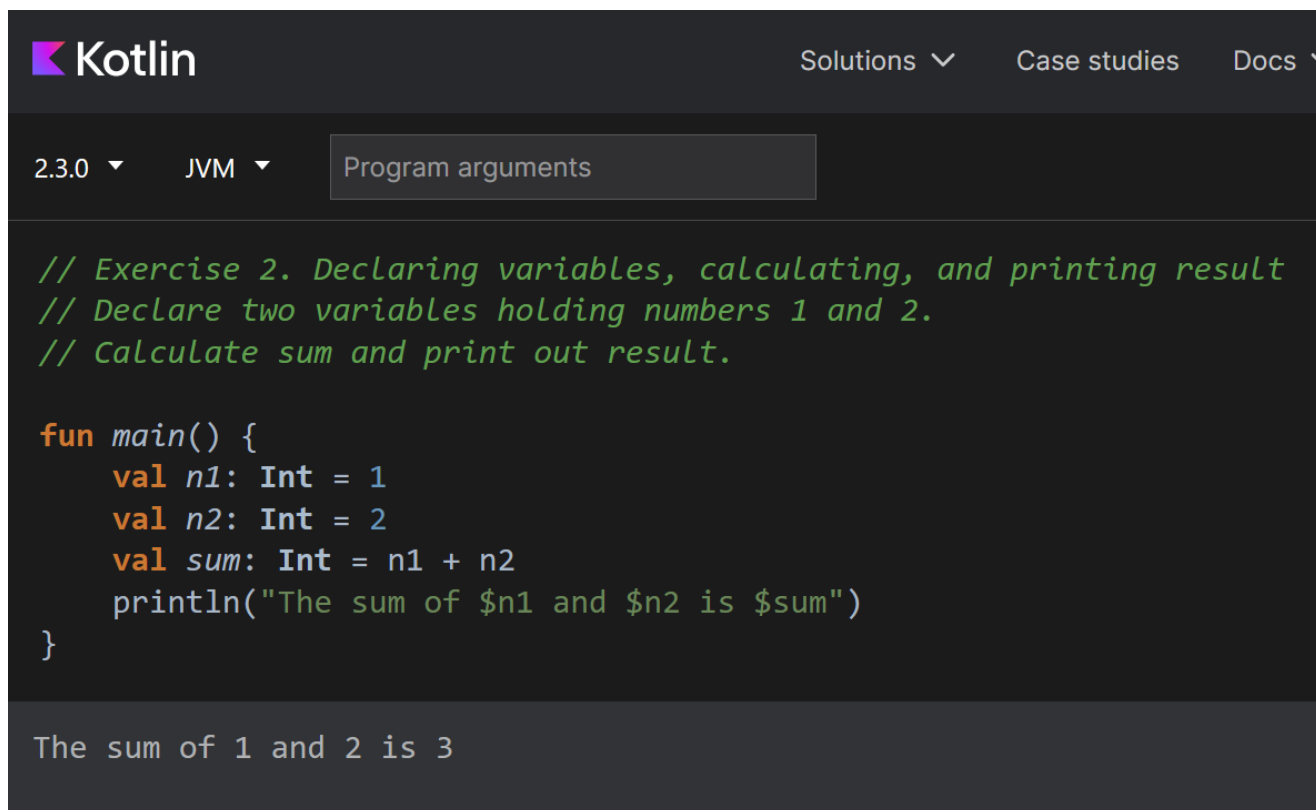
fun main() {
    val fName: String = "John"
    val lName: String = "Doe"
    println("Your name is $fName $lName")
}
```

Your name is John Doe

Figure 1. Exercise 1

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## Exercise 2



```
Kotlin Solutions Case studies Docs

2.3.0 JVM Program arguments

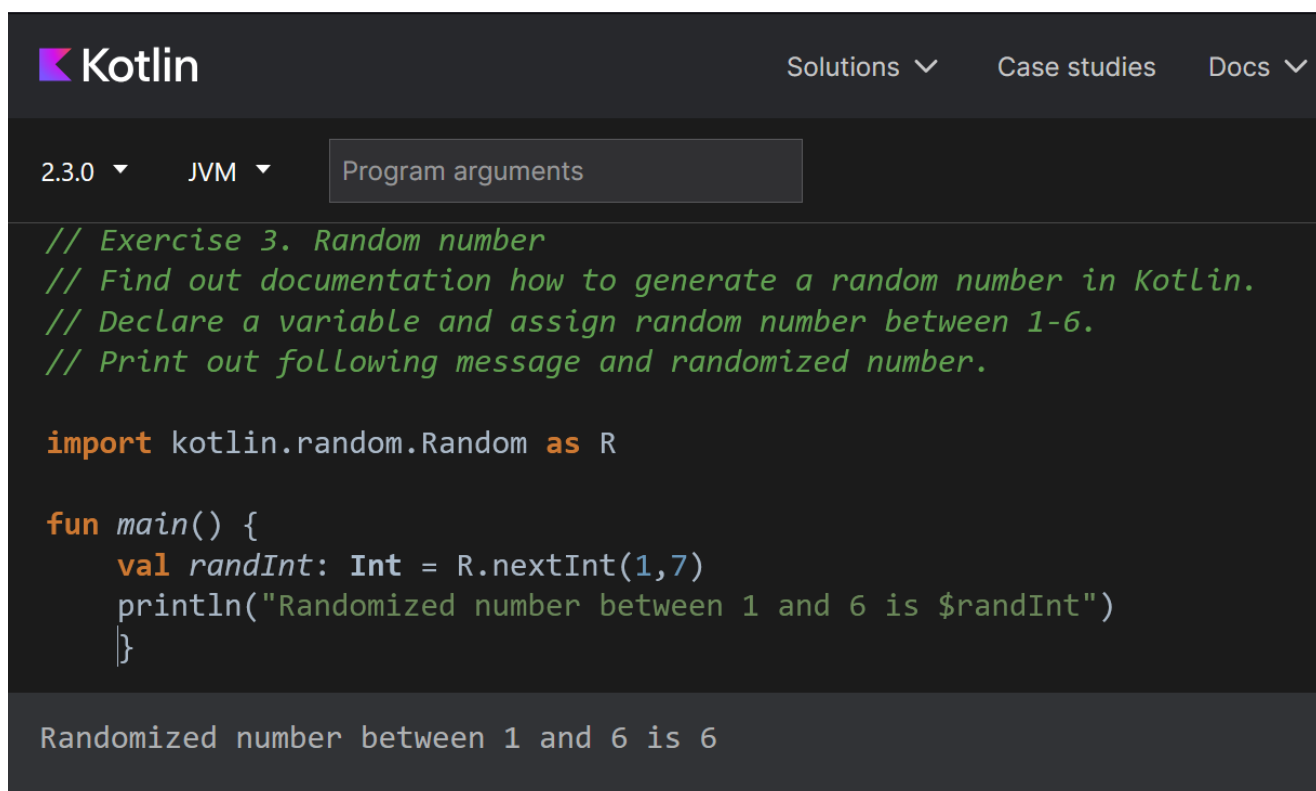
// Exercise 2. Declaring variables, calculating, and printing result
// Declare two variables holding numbers 1 and 2.
// Calculate sum and print out result.

fun main() {
    val n1: Int = 1
    val n2: Int = 2
    val sum: Int = n1 + n2
    println("The sum of $n1 and $n2 is $sum")
}

The sum of 1 and 2 is 3
```

Figure 2. Exercise 2

## Exercise 3



```
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2.3.0 JVM Program arguments

// Exercise 3. Random number
// Find out documentation how to generate a random number in Kotlin.
// Declare a variable and assign random number between 1-6.
// Print out following message and randomized number.

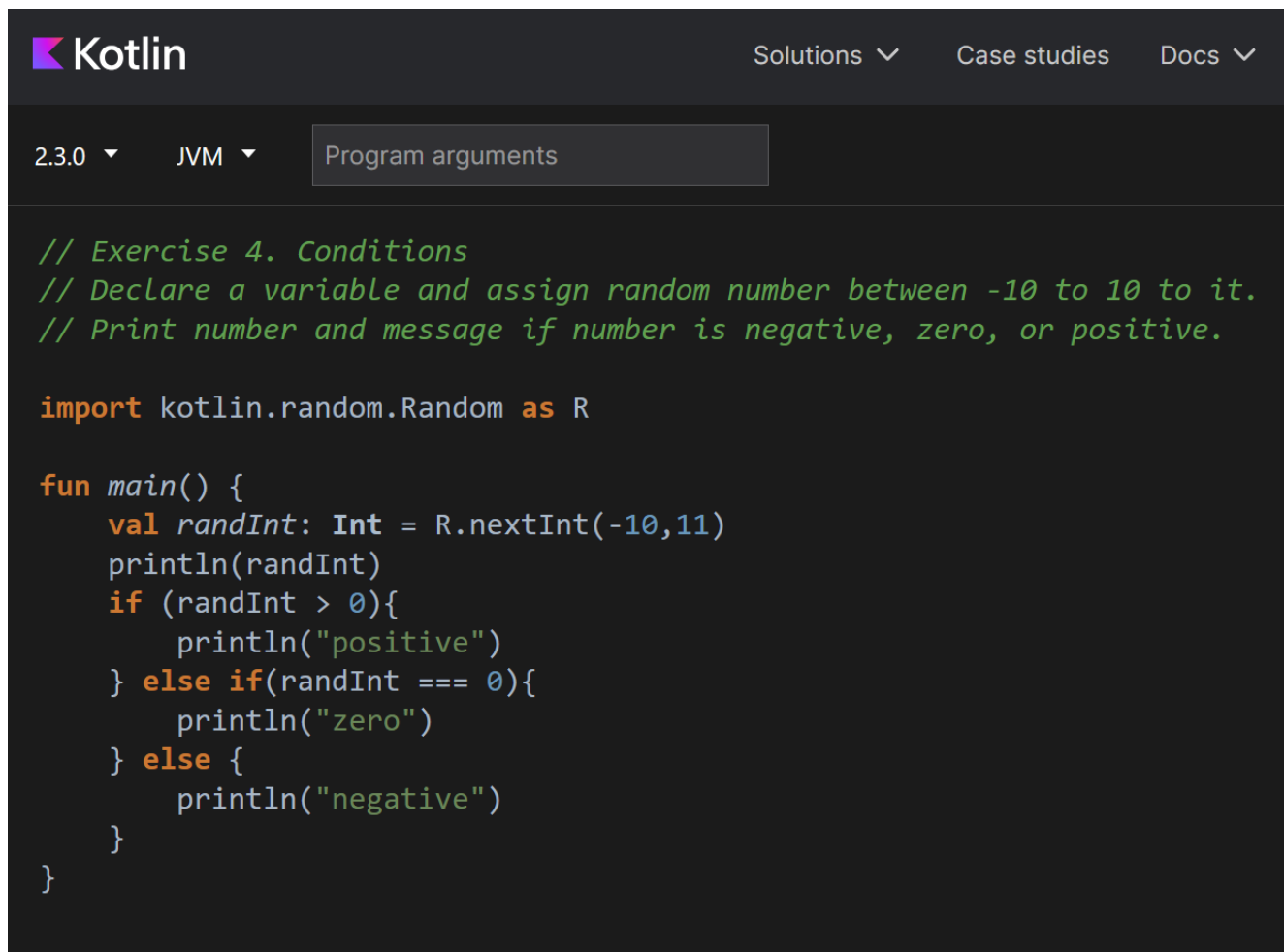
import kotlin.random.Random as R

fun main() {
    val randInt: Int = R.nextInt(1,7)
    println("Randomized number between 1 and 6 is $randInt")
}

Randomized number between 1 and 6 is 6
```

Figure 3. Exercise 3

## Exercise 4



```
Kotlin Solutions Case studies Docs

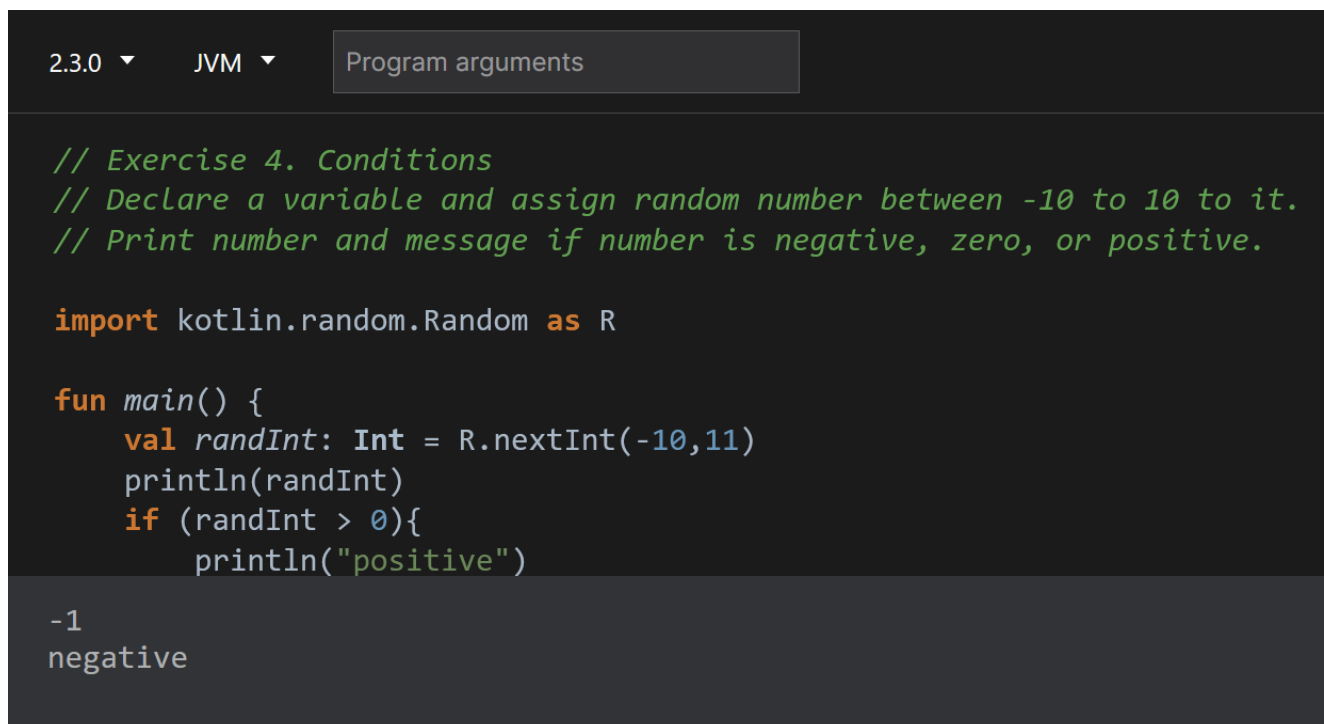
2.3.0 JVM Program arguments

// Exercise 4. Conditions
// Declare a variable and assign random number between -10 to 10 to it.
// Print number and message if number is negative, zero, or positive.

import kotlin.random.Random as R

fun main() {
    val randInt: Int = R.nextInt(-10,11)
    println(randInt)
    if (randInt > 0){
        println("positive")
    } else if(randInt === 0){
        println("zero")
    } else {
        println("negative")
    }
}
```

Figure 4. Exercise 4 - Code



```
2.3.0 JVM Program arguments

// Exercise 4. Conditions
// Declare a variable and assign random number between -10 to 10 to it.
// Print number and message if number is negative, zero, or positive.

import kotlin.random.Random as R

fun main() {
    val randInt: Int = R.nextInt(-10,11)
    println(randInt)
    if (randInt > 0){
        println("positive")
    } else if(randInt === 0){
        println("zero")
    } else {
        println("negative")
    }
}
```

-1  
negative

Figure 5. Exercise 4 - Negative

2.3.0 ▾ JVM ▾ Program arguments [Copy link](#)

```
// Exercise 4. Conditions
// Declare a variable and assign random number between -10 to 10 to it.
// Print number and message if number is negative, zero, or positive.

import kotlin.random.Random as R

fun main() {
    val randInt: Int = R.nextInt(-10,11)
    println(randInt)
    if (randInt > 0){
        println("positive")
    }
}
```

0  
zero

Figure 6. Exercise 4 - Zero

2.3.0 ▾ JVM ▾ Program arguments

```
// Exercise 4. Conditions
// Declare a variable and assign random number between -10 to 10 to it.
// Print number and message if number is negative, zero, or positive.

import kotlin.random.Random as R

fun main() {
    val randInt: Int = R.nextInt(-10,11)
    println(randInt)
    if (randInt > 0){
        println("positive")
    }
}
```

4  
positive

Figure 7. Exercise 4 - Positive

## Exercise 5

```
// Exercise 5. Arrays and loops
// Declare an array holding 5 names. Print out contents of the array using a loop.
// Randomize a number between 0 and size of the array.
// Print out a message as illustrated which contains randomized name from the array.

import kotlin.random.Random as R

fun main() {
    val names: Array<String> = arrayOf("Leonardo", "Michelangelo", "Donatello", "Raphael", "April O'Neil")
    for (name in names){
        println(name)
    }
    val winner: String = names.random()
    println("The winner is $winner")
}
```

Figure 8. Exercise 5 - Code

```
// Exercise 5. Arrays and loops
// Declare an array holding 5 names. Print out contents of the array using a loop.
// Randomize a number between 0 and size of the array.
// Print out a message as illustrated which contains randomized name from the array.

import kotlin.random.Random as R

fun main() {
    val names: Array<String> = arrayOf("Leonardo", "Michelangelo", "Donatello", "Raphael", "April O'Neil")
    for (name in names){
        println(name)
    }
    val winner: String = names.random()
    println("The winner is $winner")
}
```

Leonardo  
Michelangelo  
Donatello  
Raphael  
April O'Neil  
The winner is April O'Neil

Figure 9. Exercise 5 - Result

## Exercise 6

```
/* Exercise 6. Functions  
  Declare a function that receives age as parameter/argument and returns nothing.  
  Function calculates heart rate limits based on age and prints out the result:  
  Formulas for calculation are:  
  Lower: (220 - age) * 0.65  
  Higher: (220 - age) * 0.85  
  Call function from main and pass age as parameter (you can hardcode this).  
  Program should print out heart rate limits as illustrated.  
  */  
  
import kotlin.random.Random as R  
  
fun main() {  
    for (i in 1..3){  
        val age = (10..100).random()  
        println(age)  
        calculateHeartLimits(age)  
    }  
}  
  
fun calculateHeartLimits (age: Int) {  
    val min: Double = (220 - age) * 0.65  
    val max: Double = (220 - age) * 0.85  
    println("Heart rate limits are $min - $max")  
}
```

Figure 10. Exercise 6 - Code

```

/* Exercise 6. Functions
   Declare a function that receives age as parameter/argument and returns nothing.
   Function calculates heart rate limits based on age and prints out the result:
   Formulas for calculation are:
   Lower: (220 - age) * 0.65
   Higher: (220 - age) * 0.85
   Call function from main and pass age as parameter (you can hardcode this).
   Program should print out heart rate limits as illustrated.
   */

import kotlin.random.Random as R

fun main() {
    for (i in 1..3){
        val age = (10..100).random()
        println(age)
        calculateHeartLimits(age)
    }
}

70
Heart rate limits are 97.5 - 127.5
100
Heart rate limits are 78.0 - 102.0
19
Heart rate limits are 130.65 - 170.85

```

Figure 11. Exercise 6 - Result

## Exercise 7

```

/* Exercise 7. Arrays, Loops, functions and converting data types
   Create ArrayList of integers containing 10 random numbers.
   Print contents of ArrayList.
   Create a function that takes ArrayList of integers as parameter and returns Float
   containing calculated average for the ArrayList received as argument (there is a need
   for data type conversion since two integers are divided and result is float).
   */

import kotlin.random.Random as R

fun main() {
    val numbers = IntArray(10) {
        R.nextInt(-10, 10)
    }
    println(numbers.joinToString(", "))
    val avg = calculateAverage(numbers)
    println("The average is $avg")
}

fun calculateAverage(arr: IntArray): Double {
    val avg: Double = arr.sum().toDouble() / arr.size.toDouble()
    return avg
}

```

Figure 12. Exercise 7 - Code

```
/* Exercise 7. Arrays, Loops, functions and converting data types
Create ArrayList of integers containing 10 random numbers.
Print contents of ArrayList.
Create a function that takes ArrayList of integers as parameter and returns Float
containing calculated average for the ArrayList received as argument (there is a need
for data type conversion since two integers are divided and result is float).
*/

import kotlin.random.Random as R

fun main() {
    val numbers = IntArray(10) {
        R.nextInt(-10, 10)
    }
    println(numbers.joinToString(", "))
    val avg = calculateAverage(numbers)
    println("The average is $avg")
}

-9, -8, -7, -7, -7, -9, 8, 0, 0, -9
The average is -4.8
```

Figure 13. Exercise 7 - Result

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